

NETWORK OPERATIONS STRATEGY

MEMO

Delhivery - Graph Intelligence Initiative

Executive Summary

We analyzed 140,859 trip segments across 1,590 facilities and 2,506 corridors that make up the network. On a majority of shipments (84.8%), actual transit time deviates from the system's baseline routing estimate (OSRM) by more than 20%. This is not a reflection of SLA performance today, but it does mean our routing engine under-represents real network behavior. That gap is precisely where a graph-based view of the network creates value. By learning how delays propagate through hubs and corridors, we can produce ETAs that remain reliable under real operating conditions.

Model Impact: From Routing Estimate to Reliable ETA

Augmenting our ETA model with network structure, including hub centrality, corridor congestion patterns, and facility connectivity, improves prediction error from 10.34 minutes to 9.91 minutes, representing a 4.2% reduction in MAE. The model also increases the share of shipments with an ETA within 15% of actual delivery time from 36.9% to 41.1%, delivering a 4.2 percentage point improvement in business accuracy.

In practical terms, more shipments arrive within the promised delivery window, customer escalations decrease, and capacity planning becomes aligned with actual network behavior rather than ideal routing assumptions.

Top 5 Bottleneck Hubs: SLA Breach Contribution

1. IND000000ACB

- Monthly Breach Trips: 13,338 (82.1% of outgoing)
- Monthly Revenue Impact: ₹96.7L

- Bottleneck Score: 0.862
- Recommended Action: Parallel Route + Capacity Expansion

2. IND562132AAA

- Monthly Breach Trips: 5,965 (81.4% of outgoing)
- Monthly Revenue Impact: ₹43.2L
- Bottleneck Score: 0.659
- Recommended Action: Parallel Route + Capacity Expansion

3. IND421302AAG

- Monthly Breach Trips: 5,771 (91.2% of outgoing)
- Monthly Revenue Impact: ₹41.8L
- Bottleneck Score: 0.379
- Recommended Action: Parallel Route + Capacity Expansion

4. IND411033AAA

- Monthly Breach Trips: 2,718 (82.1% of outgoing)
- Monthly Revenue Impact: ₹19.7L
- Bottleneck Score: 0.352
- Recommended Action: Parallel Route + Capacity Expansion

5. IND501359AAE

- Monthly Breach Trips: 2,268 (89.4% of outgoing)
- Monthly Revenue Impact: ₹16.4L
- Bottleneck Score: 0.458
- Recommended Action: Parallel Route + Capacity Expansion

Corridor-Specific Interventions

A total of **1,853 chronic corridors** were identified where transit times consistently exceeded OSRM estimates by more than 20%. These corridors accounted for **83,915 breach-trips** during the analysis period.

Recommended Intervention by Corridor Profile

Long-haul corridors (>300 km) with high breach rates

- Shift from Carting to FTL
- Economic analysis indicates FTL becomes cost-neutral beyond approximately 350 km

Urban last-mile corridors operating at night

- Introduce dedicated night-shift operations at source hubs
- Reduce queueing and loading delays

Medium-distance corridors with seasonal demand spikes

- Pre-position surge capacity at the top three bottleneck hubs
- Improve resilience during peak demand periods

Quantified Impact of Upgrading the Top 3 Hubs

- Current Annual Revenue at Risk (Full Network): ₹75.3 Cr
- Annual Impact Attributable to Top 3 Hubs: ₹21.8 Cr
- Expected SLA Breach Reduction Target: ~14.5% of total network breaches
- Estimated Upgrade Capex per Hub: ₹15L
- Estimated Payback Period: Under 1 month (~0.5 months)

The upgrade investment is recovered rapidly through reduced SLA penalties and retained customer revenue, making these interventions among the highest-return operational improvements identified in this study.

- Estimated 12-Month Net Recovery (3 hubs, 50% breach reduction): ₹1046L

FTL vs Carting Optimization

Analysis indicates that **37.1% of current route assignments are sub-optimal** when evaluated using distance, breach rate, and graph-position intelligence.

Key findings:

- Estimated Monthly Savings from Graph-Informed Route Selection: ₹1966L
- FTL should generally be preferred when:
 - Distance exceeds 350 km, or
 - Corridor breach rate exceeds 35%

Implementing graph-aware route-type recommendations can significantly reduce operating

costs while improving delivery reliability.

Recommended Next Steps (0-30 Days)

1. Run the graph-enhanced ETA model in shadow mode alongside OSRM and validate performance improvements on live shipments.
2. Instrument the top three bottleneck hubs with real-time dwell-time monitoring.
3. Pilot FTL conversion on the five highest-impact Carting corridors with distances above 400 km and breach rates above 30%.
4. Deploy automated alerts for corridors exceeding a 40% breach rate threshold.

Medium-Term Roadmap (30-90 Days)

1. Initiate facility upgrade planning for the top two hubs, prioritized by expected payback.
2. Move the graph-enhanced ETA model from shadow mode into a controlled operational pilot on selected corridors. Validate performance against OSRM before expanding coverage and proceed to phased production deployment only after sustained performance gains are observed.
3. Integrate the route-type recommendation engine into the dispatch workflow, with an estimated quarterly savings potential of ₹5897L.
4. Assess parallel-route feasibility for the highest-betweenness hub in the network.

Conclusion

The analysis demonstrates that logistics performance is influenced not only by distance and routing, but also by the structural position of facilities within the network. By incorporating graph intelligence into ETA prediction, bottleneck identification, and route-type optimization, Delhivery can improve delivery reliability, reduce operational inefficiencies, and recover significant revenue currently at risk due to network delays.

The recommended interventions prioritize the highest-impact hubs and corridors, offering rapid payback and measurable improvements in SLA performance, customer experience, and network efficiency.